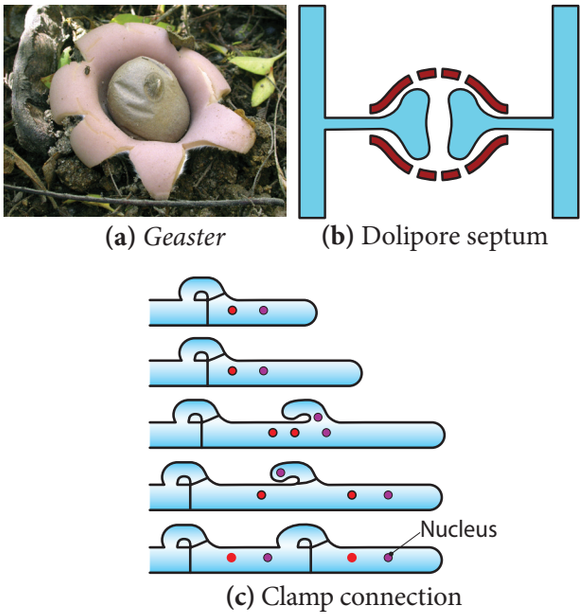
- Asexual reproduction takes place by fission, budding, oidia, conidia, and chlamydospore.
- Sexual reproduction takes place by the fusion of two compatible nuclei.
- Plasmogamy is not immediately followed by karyogamy, instead a dikaryotic condition is prolonged for several generations.
- A special hyphae called ascogenous hyphae is formed.
- A crozier is formed when the tip of the ascogenous hyphae recurves forming a hooked cell. The two nuclei in the penultimate cell of the hypha fuse to form a diploid nucleus. This cell forms young ascus.
- The diploid nucleus undergo meiotic division to produce four haploid nuclei, which further divide mitotically to form eight nuclei. The nucleus gets organised into 8 ascospores.
- The ascospores are found inside a bag like structure called ascus. Due to the presence of ascus, this group is popularly called "Sac fungi".
- Asci gets surrounded by sterile hyphae forming fruit body called ascocarp.
- There are 4 types of ascocarps namely **Cleistothecium** (Completely closed), **Perithecium** (Flask shaped with ostiole), **Apothecium** (Cup shaped, open type) and **Pseudothecium**.

Basidiomycetes

- Basidiomycetes include puff balls, toad stools, Bird's nest fungi, Bracket fungi, stink horns, rusts and smuts (Figure 1.22).
- The members are terrestrial and lead a saprophytic and parasitic mode of life.
- The mycelium is well developed, septate with dolipore septum(bracket like). Three types of mycelium namely primary (Monokaryotic), secondary (Dikaryotic) and tertiary are found.



(a) *Geaster* **(b)** Dolipore septum

(c) Clamp connection

Figure 1.22: Structure and Reproduction in Basidiomycetes

- Clamp connections are formed to maintain dikaryotic condition.
- Asexual reproduction is by means of conidia, oidia or budding.
- Sexual reproduction is present but sex organs are absent. Somatogamy or spermatisation results in plasmogamy. Karyogamy is delayed and dikaryotic phase is prolonged. Karyogamy takes place in basidium and it is immediately followed by meiotic division.
- The four nuclei thus formed are transformed into basidiospores which are borne on sterigmata outside the basidium (Exogenous). The basidium is club shaped with four basidiospores, thus this group of fungi is popularly called "Club fungi". The fruit body formed is called Basidiocarp**.**

Deuteromycetes or Fungi Imperfecti

The fungi belonging to this group lack sexual reproduction and are called imperfect fungi. A large number of species live as saprophytes in soil and many are plant and animal parasites. Asexual reproduction takes place by the production of conidia, chlamydospores, budding, oidia etc.,

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